Doudou GT-U7

Doudou GT-U7 GPS Module User Manual

High Sensitivity, Low Power Consumption GPS Module

1. Introduction

The Doudou GT-U7 GPS Module is a compact and highly sensitive positioning device designed for a wide range of applications requiring accurate location data. Its low power consumption and miniaturized design make it ideal for integration into various systems, including car navigation, drones, vehicle monitoring, and mobile devices. This module is compatible with popular platforms such as NEO-6M and STM32.

The GT-U7 module excels in environments where ordinary GPS receivers struggle, such as urban areas with narrow sky views or dense jungle environments, due to its extremely high tracking sensitivity. It offers stable performance with minimal static drift.

Image: Two GT-U7 GPS modules, each accompanied by its IPX antenna and a 5-pin header, illustrating the complete package contents.

2. Product Features

- **High Sensitivity:** Enables accurate positioning even in challenging environments like narrow urban canyons or dense foliage.
- **Low Power Consumption:** Designed for energy efficiency, extending battery life in portable applications.
- **Miniaturization:** Compact size (27.6x26.6mm) with positioning holes for easy integration.

- **Wide Compatibility:** Fully compatible with NEO-6M, 51 MCU, and STM32 platforms.
- **Flexible Power Supply:** Operates on 3.6V-5V or can be powered directly via Micro USB.
- **Active Ceramic Antenna:** Includes an IPEX interface and an active ceramic antenna for rapid satellite acquisition and positioning.
- **LED Signal Indication:** Onboard LED provides visual feedback on signal status.
- **Data Backup Battery:** Integrated battery for maintaining data and faster hot starts.
- **Onboard E2PROM:** Allows for saving and retaining parameter data.
- **Standard Output:** NEMA output format for broad compatibility with GPS software and systems.

3. Package Contents

Each package of the Doudou GT-U7 GPS Module typically includes the following components:

- GT-U7 GPS module (x1)
- IPX antenna (x1)
- 5pin 2.54mm pin header (x1)

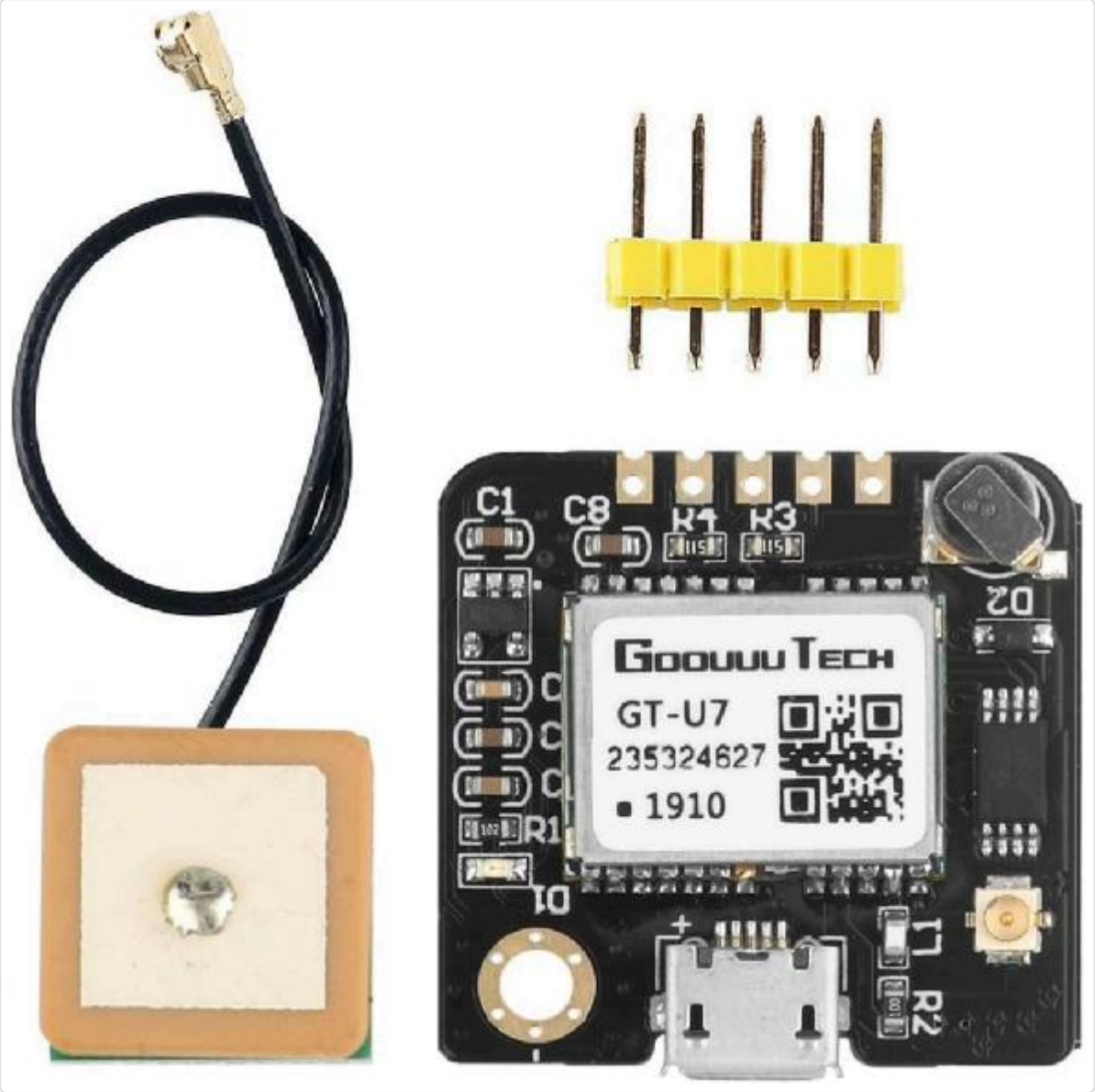

Image: A single GT-U7 GPS module, its IPX antenna, and the included 5-pin header, detailing the individual components.

4. Setup Guide

Follow these steps to set up your GT-U7 GPS Module:

1. **Power Supply:** The module can be powered via its 2.54mm pitch in-line connection interface (VCC pin) with a voltage range of 3.6V to 5V. Alternatively, you can use a Micro USB cable to directly supply power to the module. Ensure the power source provides stable voltage within the specified range.

2. **Antenna Connection:** Connect the provided IPX active ceramic antenna to the IPEX antenna interface on the module. Ensure a secure connection for optimal signal reception.
3. **Pin Connections:** Utilize the 2.54mm pitch in-line connection interface for data communication. The common pins are:
    - **VCC:** Power input (3.6V-5V)
    - **GND:** Ground
    - **TXD:** Transmit Data (output from module)
    - **RXD:** Receive Data (input to module)
    - **PPS:** Pulse Per Second (timing signal output)
4. **Integration:** Mount the module using its positioning holes if integrating into a larger system. Ensure proper ventilation and avoid placing it near strong electromagnetic interference sources.

Image: The reverse side of the GT-U7 GPS module, clearly labeling the PPS, TXD, RXD, GND, and VCC pins for easy connection.

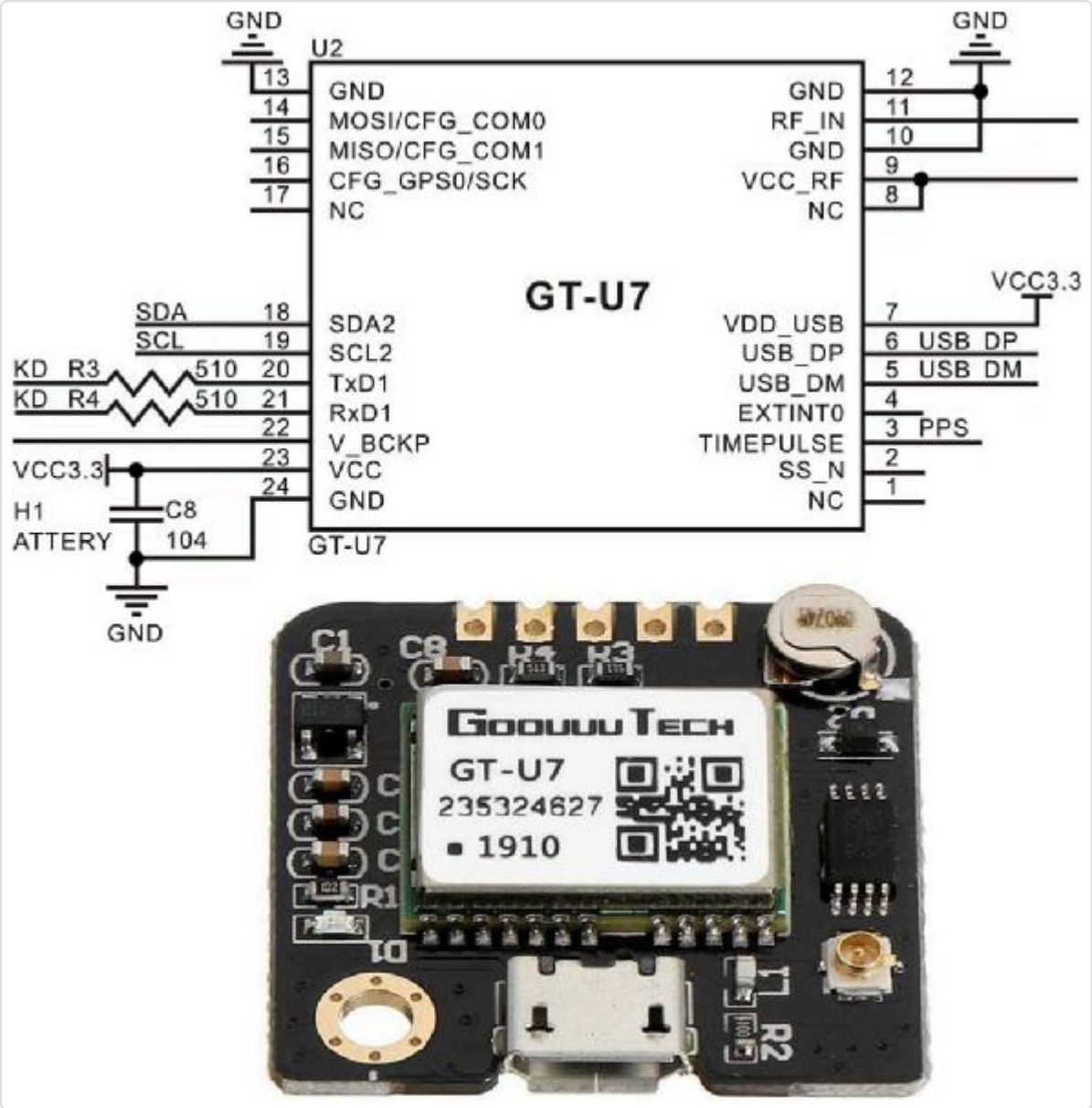

Image: A detailed schematic diagram of the GT-U7 GPS module, illustrating its internal connections and pinout for advanced users.

5. Operating Instructions

Once the GT-U7 module is powered and connected, it will begin to acquire satellite signals. The onboard LED will indicate signal status (e.g., blinking for acquisition, solid for fix).

- **Baud Rate:** The default working baud rate for the module is 9600 bps. This can be modified programmatically if required by your application.

- **NEMA Output:** The module outputs positioning data in the standard NEMA format, which is widely supported by GPS parsing libraries and software. This format is compatible with NEO-6M modules.
- **Data Backup:** The onboard E2PROM allows the module to save configuration and parameter data, ensuring that settings are retained even after power cycles.
- **Positioning:** For optimal performance, ensure the antenna has a clear view of the sky. The module's high sensitivity allows it to function even in partially obstructed environments.

Image: The GT-U7 GPS module and its antenna positioned next to a smartphone displaying a map, illustrating a typical application scenario for location tracking.

6. Specifications

Parameter	Value
Model Name	GT-U7
Working Voltage	3.6V-5V (or Micro USB)
Working Baud Rate	9600 (default, modifiable)
Module Dimensions (LxWxH)	27.6mm x 26.6mm x ~1.05" (approx. 26.6mm)
Antenna Interface	IPEX
Antenna Dimensions	15mm x 15mm x 6mm (approx.)
Output Format	NEMA (compatible with NEO-6M)
Onboard Features	LED signal indication, Data backup battery, E2PROM
Connectivity Technology	USB
Mounting Type	Panel Mount
Item Weight	0.634 ounces

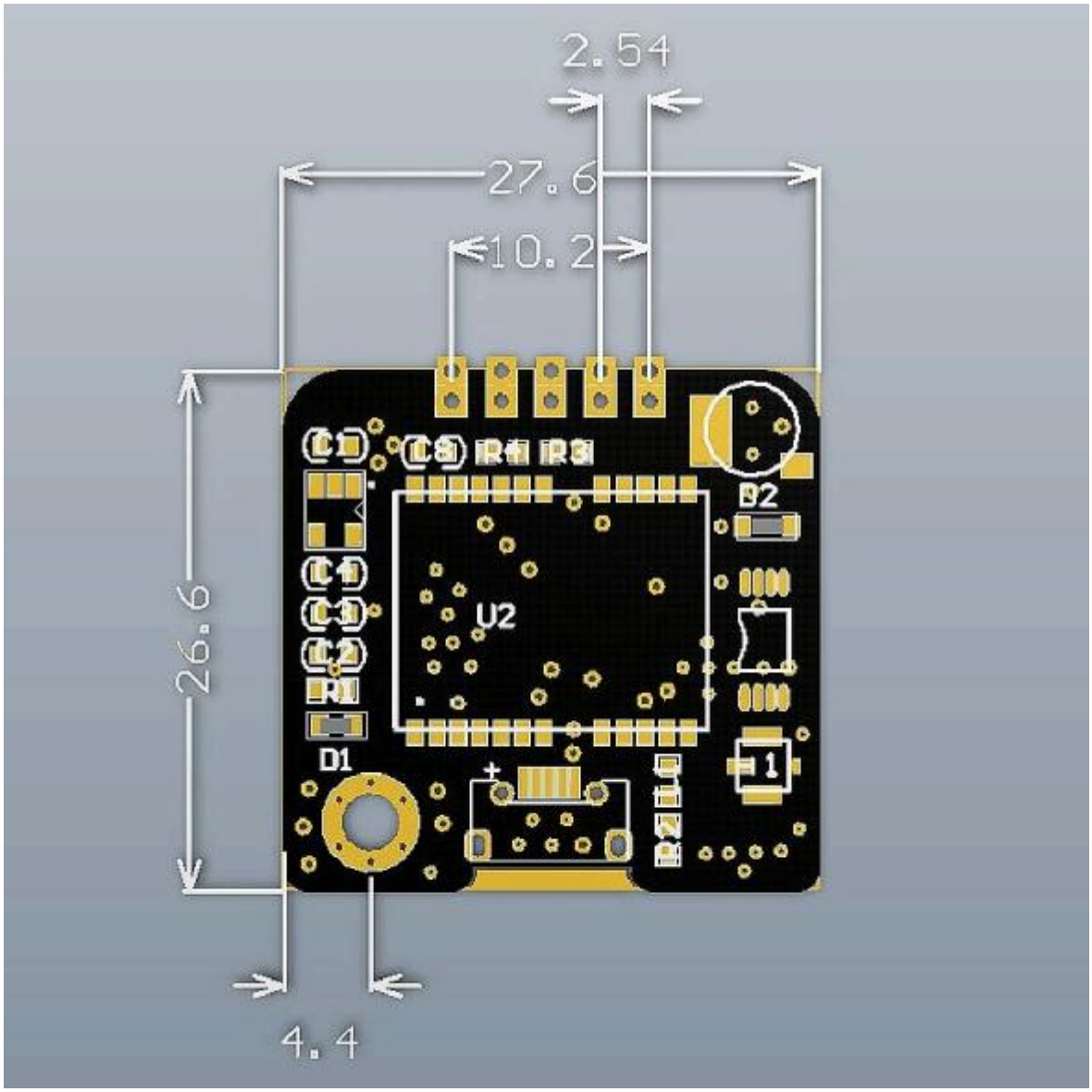

Image: A technical drawing illustrating the precise dimensions of the GT-U7 GPS module in millimeters, crucial for integration into custom projects.

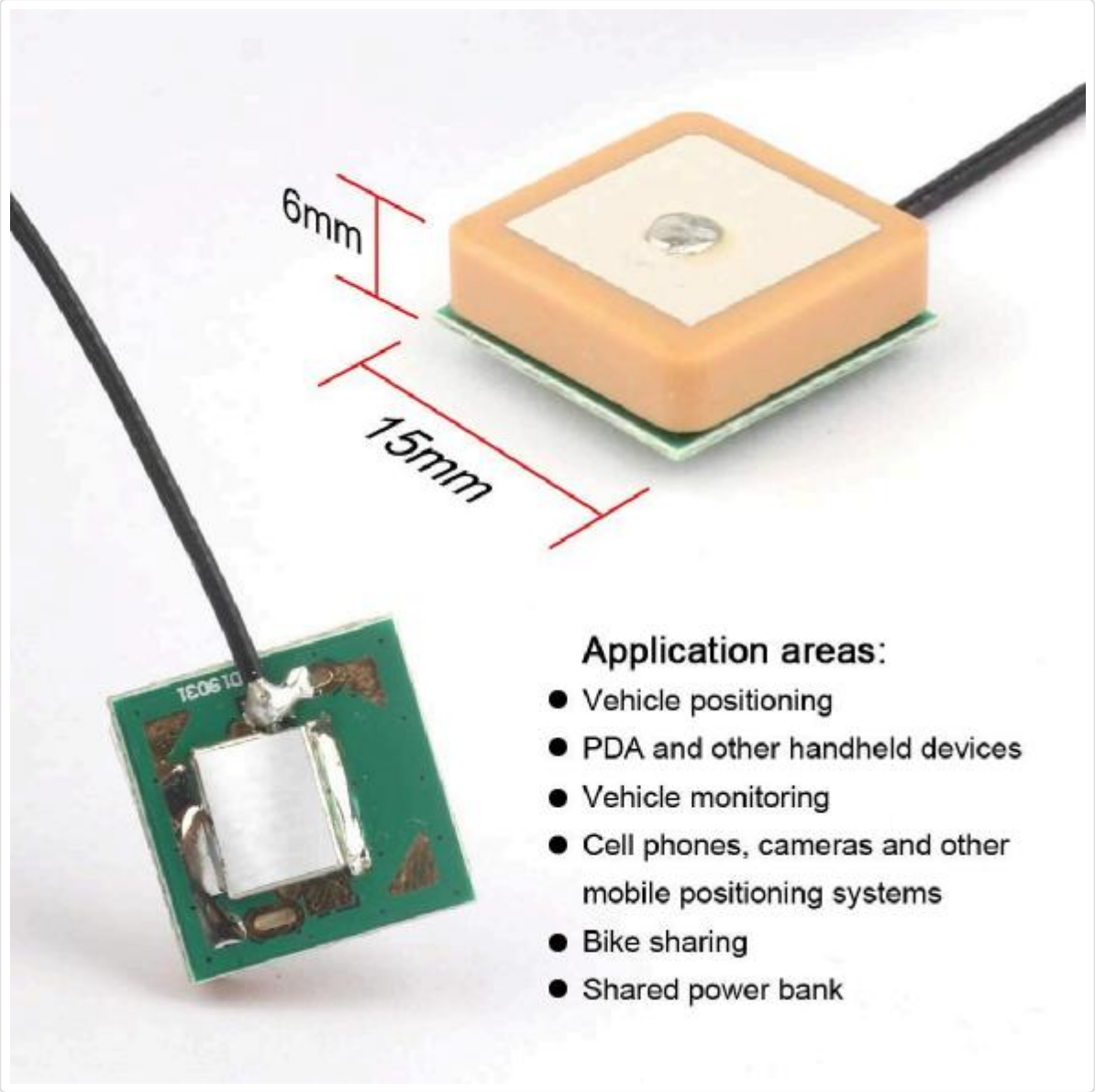

Image: This image displays the dimensions of the IPX antenna (6mm height, 15mm width) and lists various application areas for the GT-U7 module.

7. Application Areas

The Doudou GT-U7 GPS Module is suitable for a wide range of positioning applications due to its high sensitivity and compact size. Key application areas include:

- Car navigation systems
- Drones and UAVs
- Vehicle monitoring and tracking

- Shared bicycles and e-scooters
- Mobile phones and portable devices requiring GPS
- Cameras with geotagging capabilities
- PDA and other handheld devices
- Shared power banks

8. Maintenance

To ensure the longevity and optimal performance of your GT-U7 GPS Module, consider the following maintenance guidelines:

- **Keep Dry:** Protect the module from moisture and humidity, which can damage electronic components.
- **Avoid Extreme Temperatures:** Operate and store the module within its specified temperature ranges to prevent damage.
- **Clean Gently:** If cleaning is necessary, use a soft, dry cloth. Avoid using harsh chemicals or abrasive materials.
- **Handle with Care:** Electronic components are delicate. Avoid dropping the module or subjecting it to strong impacts.
- **Secure Connections:** Periodically check that all pin and antenna connections are secure and free from corrosion.

9. Troubleshooting

If you encounter issues with your GT-U7 GPS Module, consider the following troubleshooting steps:

- **No GPS Signal:**
    - Ensure the antenna is properly connected to the IPEX interface.
    - Verify the antenna has a clear, unobstructed view of the sky.

    - Check the LED indicator on the module; if it's not blinking or solid, it may not be acquiring satellites.
- **Incorrect or No Data Output:**
    - Confirm the power supply voltage is within the 3.6V-5V range.
    - Verify that the TXD and RXD pins are correctly connected to your microcontroller or receiving device.
    - Ensure your receiving device's serial port baud rate is set to 9600 bps (or matches the module's configured baud rate).
    - Check for any short circuits or loose connections on the module or connecting wires.
- **Module Not Powering On:**
    - Confirm the power supply is active and providing the correct voltage.
    - If using Micro USB, try a different cable or power source.
    - Ensure the VCC and GND pins are correctly wired.

10. Warranty and Support

Specific warranty details for the Doudou GT-U7 GPS Module are not provided in the product information. For warranty claims, technical support, or further inquiries, please refer to the seller or manufacturer's official contact channels. It is recommended to retain your purchase receipt for any potential support needs.

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Product Model: GT-U7

Related Documents - GT-U7

[Black Shark GT Neo Smartwatch User Manual]

Comprehensive user manual for the Black Shark GT Neo smartwatch, covering setup, features, charging, strap replacement, and safety information.

[Black Shark GT Neo Smartwatch User Manual and Setup Guide]

Comprehensive guide to setting up, using, and maintaining your Black Shark GT Neo smartwatch, including features, specifications, and safety warnings.